

Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

Peter Cotton*

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Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. We measure how much that omission matters. Across fifteen named, seeded ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.07 to 0.86, and eleven audited claims from eleven fields flip verdict as the ensemble changes, for stated structural reasons. The `randomcov` library names the measures and seeds the draws, so any claim can be re-run under all fifteen; the results assemble into a matrix of claims against measures, and a joint leaderboard of scikit-learn and `precise` estimators has five different winners. The pattern survives scale: six of seven claims re-run from $n = 30$ to $n = 3000$ keep their verdict, and the seventh, hierarchical risk parity, flips under its own home-turf ensemble as dimension grows. Eleven audits are completed here; a docket from astronomy, genomics, neuroscience, and chemometrics, including ensemble-invariant null controls, is assembled for the rest.

1 The unstated measure

Empirical claims about covariance methods have the form “estimator A beats estimator B ,” “the shrinkage intensity is typically moderate,” “the factorization recovers structure,” with performance averaged over random test matrices. That average is of course taken with respect to some generating law, and in practice the law is usually inherited from convenience, and from the field. Finance validates on equity panels; econophysics on the same panels with random-matrix nulls; geostatistics on smooth kernels; psychometrics on low-rank factor models; graphical modeling on sparse precisions; cosmology on mock catalogs of ordered correlation-function bins.

Different laws produce matrices with different spectra, different locality, and different factor content. So it should be no surprise that a verdict certified under one law carries no warrant under another. What the measurements below show is that this omission is not a mere technicality.

*`peter.cotton@microprediction.com`. The ensemble library is `pip install randomcov`; the seeded scripts behind every number here live beside this paper at `github.com/microprediction/randomcov`.

2 Precedents

We are hardly the first to notice that method comparisons inherit their verdicts from the simulation design; this is established already in several literatures. Morris et al. [2019] require simulation studies to declare the data-generating mechanism as a design choice of equal rank with the estimand. Boulesteix et al. [2013] trace the over-optimism of self-reported comparisons to test beds chosen by the method’s own authors. Nießl et al. [2022] re-analyze one benchmark under hundreds of design and analysis options and find that almost any method can reach almost any rank. Hooker [1995] made the complaint for heuristics; Wolpert and Macready [1997] give the limit statement, that averaged over all objectives all optimizers tie.

The correlation-matrix version also has precedent, which is reassuring. Joe [2006] and Lewandowski et al. [2009] supply the standard samplers. Hardin et al. [2013] argue that the habitual generators produce unrealistic matrices and bias the evaluation of statistical techniques, and propose a better one. Marti [2020] learns a generator from financial correlation matrices and reports, in accompanying experiments, that the HRP advantage fades on matrices without hierarchical structure, a result restated in Marti et al. [2021]. Papenbrock et al. [2021] evolve synthetic correlations to ask which correlation structures favor which allocator. Alvarez et al. [2014] document the Bayesian mirror image: an inverse-Wishart prior biases the posterior variance upward and the correlation toward zero when the true variance is small relative to the prior mean, and the bias persists as the sample grows.

What the present paper adds is the instrument and the audits: a fixed battery of fifteen named, seeded, public measures; the re-execution of eleven published claims from eleven fields under all of them; the claims-by-measures matrix as the reporting format; and provably invariant theorems run as null controls.

3 Fifteen named ensembles

`randomcov` generates correlation matrices under fifteen named, seeded measures, and we tried to keep the naming honest rather than cute: `lkj`, `onion`, `vine` (elliptope samplers; Joe, 2006, Lewandowski et al., 2009); `wishart`, `spectrum` (spectral laws, including Marchenko–Pastur and spiked); `factor`, `hierarchical`, `block_equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse_precision` (graphical-model structure); `archakov_hansen` (log-matrix parameterization; Archakov and Hansen, 2021); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles).

Variance vectors are drawn separately, so correlation structure and scale can be crossed independently. Every generator accepts an explicit seed, and every number in this paper descends from one – there is no number here we cannot regenerate. The scripts sit beside the paper.

4 Demonstration: one estimator, twelve-fold spread

We begin by looking at the well known Ledoit-Wolf shrinkage estimator because this is used across so many applications. Moreover the diagnostic is easy to understand and states how far the sample covariance should be pulled towards a “target” (using neutral terminology – the target can take different flavors). The table below demonstrates the variation. Here we are fixing $n = 30$ variables each observed exactly $T = 60$ times. Even though this is not pressing

up against the critical $n = T$ boundary we still see tremendous variation. We used 20 draws to get a quick sense.

ensemble	intensity	sd
residuals	0.072	(0.019)
animals	0.072	(0.013)
factor	0.101	(0.038)
archakov_hansen	0.120	(0.022)
kernel	0.132	(0.082)
walk	0.199	(0.038)
block_equicorr	0.247	(0.096)
vine	0.355	(0.022)
lkj	0.363	(0.024)
onion	0.363	(0.030)
hierarchical	0.369	(0.070)
ar1	0.469	(0.226)
wishart	0.506	(0.032)
spectrum	0.532	(0.058)
sparse_precision	0.862	(0.066)

The intensity spans 0.072 to 0.862 at identical (n, T) . So clearly, any paper reporting “typical” shrinkage from one generator is actually reporting a property of its generator. The ordering is interpretable too. We see that strong low-rank structure (**animals**, **residuals**, **factor**) wants little shrinkage, and near-unstructured draws (**wishart**, **sparse_precision**) want a great deal. The within-ensemble standard deviation is itself informative. Note that **ar1** spans 0.23 across its own parameter draws.

5 Demonstration: one algorithm, one failing ensemble

We pick on one of our own examples. A deterministic algorithm for multinomial probit choice probabilities (the **winning** package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. Often this is quite good. In fact for twelve of the fifteen measures the error is $1\text{--}4 \times 10^{-5}$. On the other hand for what we term dense-strong ensembles it is $1\text{--}4 \times 10^{-4}$ and under **kernel** fields (Matérn and RBF) it is 4×10^{-3} . In this case this is predictable because the fit in question does not yet admit band-diagonal structure catering to kernel covariance patterns. The point is that a validation run on any single ensemble would have reported either “essentially exact” or “inadequate,” both correctly, for the same method. Writing the random covariance generation family was a big part of the motivation: producing better software.

6 Eleven audits, eleven fields

Curiosity led us to wonder if the same brittleness of claim might apply more broadly, or whether, to put a more positive spin on the topic, there may be many opportunities to identify improvements to existing tools. Our methodology is very simple and short here. We take a published claim, re-run its demonstration using every type of covariance generator at our disposal, and report the verdict. If anything the selection was biased towards things we either

knew to be helpful or hoped would be – what that means for how the results should be read is taken up properly in Scope below.

We tried to spread this out and audit one claim from each of eleven different fields, seven about estimators and constructions (Table 1) and four about inference and dimensionality (Table 2). This part of the selection was rather ad hoc, as noted. Nonetheless Figure 1 assembles the results into a single matrix and outlines each claim’s *home turf*, by which we mean the correlation structure it was demonstrated on or intended for as best we interpret. All runs use $n = 30$ with twenty replications unless stated. Sampled panels use $T = 60$ except where the claim’s own regime dictates otherwise.

Finance. We start here because it is the field we know best. López de Prado [2016] reports that hierarchical risk parity delivers lower out-of-sample variance than a long-only, fully-invested quadratic minimum-variance optimizer, and names estimation error in the inverse as the culprit. We build both portfolios from the same sample covariance and score the true variance $w^\top \Sigma w$ under the generating covariance. Column 1 of Table 1 gives the median ratio of HRP to long-only minimum variance. HRP wins on every draw under **sparse_precision** and splits draws roughly evenly under **ar1** and **hierarchical**, the ensembles whose dependency structure the bisection tree can match; it loses by an order of magnitude or more under **animals**, **factor**, and **archakov_hansen**, where the true matrix rewards positions a long-only tree simply cannot take. Against unconstrained minimum variance the losses grow by many more orders of magnitude: the median ratio is $\sim 10^{8.7}$ under **walk** and $\sim 10^{11.2}$ under **residuals**, because near-singular ensembles admit near-perfect hedges that only signed weights can reach. Both of those ensembles reach the positive-semidefinite floor, so the exponent is partly a statement about the floor: it scales inversely with the clip, and a looser clip moves it down decade for decade. The direction is a property of the ensemble; the magnitude is not.

Econophysics. Laloux et al. [1999] find the bulk of the sample correlation spectrum indistinguishable from Marchenko–Pastur noise; the cleaning recipe that follows — flatten every eigenvalue below the edge $(1 + \sqrt{n/T})^2$ and report a cleaner matrix — is Laloux et al. [2000]. Column 2 gives the Frobenius-loss ratio of clipped to raw sample correlation. We observe that this “cleaning” is excellent under **sparse_precision** (0.30) and **hierarchical** (0.76). The mechanism is not quite the textbook one. **hierarchical** does carry a spike over a bulk, but **sparse_precision** carries none — its largest true eigenvalue is 1.43 against an edge of 2.91 — and clipping wins there because the sub-edge spectrum is nearly flat already, so flattening it costs almost nothing. What predicts the verdict is the dispersion of the true spectrum below the edge, not the presence of spikes above it. On the other hand clipping damages the estimate under eleven of fifteen ensembles, worst at 1.43–1.77 under **animals**, **kernel**, **archakov_hansen**, and **residuals**, where the sub-edge mass is dispersed and is erased as noise. The recipe presumes signal means spikes. However in contrast, locality puts signal in the bulk. Picking the right tool for the right task is important.

Geostatistics. The screening effect of Stein [2002] says that a few nearest neighbors, here the most correlated, nearly attain the full kriging variance. This is a population property, so no sampling is involved. Column 3 reports the median ratio of conditional variance given the best five predictors to conditional variance given all $n - 1$. Screening holds essentially exactly

under **ar1** (Markov, ratio 1.00), **hierarchical**, and **block_equicorr**. It also reads 1.00 under **sparse_precision**, but vacuously: conditioning on all $n - 1$ neighbors removes only about four per cent of the variance there, so there is nothing for five neighbors to miss. It fails several-fold under the elliptope samplers **lkj**, **onion** and **vine**, and under **kernel** itself (5.7). Every **kernel** draw is squared-exponential, whose spectral density decays faster than any algebraic rate and so falls outside Stein’s conditions; interpolation is nearly deterministic from all sites but not from five. The failure tracks the drawn length scale almost perfectly (correlation 0.98 between length scale and log ratio), and a Matérn-3/2 field on the identical point clouds gives 1.04 instead — which is Stein’s caveat, reproduced as a contrast rather than assumed. Thus we see that Stein’s own caveat, that screening depends on smoothness, plays out.

Psychometrics. Parallel analysis [Horn, 1965] retains the components whose sample eigenvalues exceed those of independent data of the same shape — Horn compared against the mean null eigenvalue, and we use the now-standard 95th-percentile form of Glorfeld [1995]; it is the standard answer to “how many factors?” Column 4 gives the mean retained count: 1.8 under **sparse_precision** to 7.3 under **onion** at identical (n, T) — more than a fourfold spread for what is supposed to be an objective count. The same sweep scores the eigenvalue-greater-than-one rule of Kaiser [1960] against the population count of eigenvalues above one. Kaiser over-extracts under **factor** (5.7 retained against 3.0 true), under-extracts under **wishart** (11.2 against 13.1), and disagrees with parallel analysis by a factor that is itself ensemble-dependent: 1.4 under **residuals**, 7 under **sparse_precision**. So the number of factors a study reports is, in part, a property of the generating measure its data happen to resemble.

Data assimilation. Ensemble Kalman filtering runs member-starved, so to speak, and the standard way of dealing with this is covariance localization. In other words we taper the sample covariance entrywise with a Gaspari–Cohn function [Gaspari and Cohn, 1999] of inter-variable distance [Houtekamer and Mitchell, 2001, Hamill et al., 2001]. Paz and Sánchez [2015] apply the closely related covariance tapering of Kaufman et al. [2008] to correlation-function bins, with a Wendland taper and a two-taper precision estimator; the assumption both share — that the taper coordinate carries the truth’s locality — is what this audit tests, so it applies to both fields. With $N = 20$ members for $n = 30$ variables and a taper of index distance with half-radius 6, column 5 reports the Frobenius-loss ratio of tapered to raw. The split is interpretable. Localization helps wherever the estimate is noisy and the truth carries no long-range structure in taper coordinates (including the dense elliptope samplers) where variance reduction beats the bias. However this actually hurts under **factor**, **residuals**, and **animals** (1.66–1.71), whose common factors are long-range by construction. It also reduces performance under **kernel** (1.46), the one ensemble that genuinely is local, because kernel locality lives in a latent two-dimensional point cloud rather than in index order. Localization requires locality in the coordinates the taper can see.

Signal processing. Diagonal loading is the standard repair for the sample-covariance MVDR beamformer: Cox et al. [1987] derive it from a constraint on white-noise array gain, against sensor and steering perturbations, and Carlson [1988] tie it to covariance estimation error in the snapshot-limited regime. With $T = 40$ snapshots, a random unit steering vector, the distortionless constraint, and loading $\delta = 0.1 \operatorname{tr}(\hat{\Sigma})/n$, column 6 reports the median ratio of true output power, loaded over unloaded. The picture is genuinely mixed: loading halves the

ensemble	HRP/MV (fin.)	clip/raw (econo.)	near/full (geost.)	PA count (psych.)	taper/raw (assim.)	load/raw (sig. proc.)	gl/LW (graph.)
animals	8.38	1.43	5.53	3.3	1.66	32.8	1.00
ar1	0.98	1.20	1.00	4.5	0.53	0.48	0.69
archakov_hansen	71.8	1.47	131	4.7	1.41	20.5	1.00
block_equicorr	1.09	1.00	1.04	3.0	1.00	0.55	0.78
factor	12.3	1.38	1.06	2.2	1.69	0.94	1.09
hierarchical	1.00	0.76	1.02	2.3	0.82	0.45	0.68
kernel	1.51	1.46	5.69	4.1	1.46	136	1.00
lkj	2.67	1.29	10.6	7.3	0.79	2.26	0.99
onion	2.83	1.29	9.04	7.3	0.78	3.29	1.00
residuals	2.75	1.77	$\sim 10^{8.3}$	4.0	1.84	$\sim 10^{7.7}$	1.00
sparse_precision	0.76	0.30	1.00	1.8	0.50	0.48	0.63
spectrum	1.11	0.94	1.32	5.8	0.66	0.51	0.99
vine	2.68	1.29	11.9	7.0	0.81	4.13	1.00
walk	1.23	1.11	$\sim 10^{8.7}$	3.7	1.14	$\sim 10^{7.9}$	1.00
wishart	1.19	1.04	1.50	6.5	0.66	0.51	0.95

Table 1: Seven audited claims about estimators and constructions; all entries are medians over twenty seeded draws. Column 1: out-of-sample true-variance ratio, HRP over long-only minimum variance (below one: HRP wins). Column 2: Frobenius-loss ratio, Marchenko–Pastur-clipped over raw sample correlation (below one: cleaning helps). Column 3: population conditional-variance ratio, five best predictors over all (near one: screening holds). Column 4: mean components retained by parallel analysis. Column 5: Frobenius-loss ratio, Gaspari–Cohn-tapered over raw sample covariance at $N = 20$ members (below one: localization helps). Column 6: true beamformer output power, diagonally loaded over unloaded at $T = 40$ snapshots (below one: loading helps). Column 7: precision-matrix Frobenius-loss ratio, cross-validated graphical lasso over inverted Ledoit–Wolf (below one: the sparsity prior wins). Each column’s verdict flips across rows.

output power under half the ensembles, which is the win it is sold as. But it multiplies the power under `archakov_hansen` (21), `animals` (33), `kernel` (136), and `walk` ($10^{7.9}$), which is not a small miss. The mechanism is the hedge from the finance audit seen from the other side: under near-singular ensembles the beamformer’s null-steering lives in the smallest eigenvalues, which is exactly what loading erases.

Graphical modeling. The graphical lasso of Friedman et al. [2008] made the sparsity prior on the precision matrix cheap enough to be routine — their own claim is speed, not accuracy — and the field attached to it the promise that the prior improves estimation. That promise underwrites partial-correlation connectomes [Smith et al., 2011] and sparse cosmological precision matrices [Padmanabhan et al., 2016]. Column 7 reports the Frobenius loss of the cross-validated glasso precision against the true precision, relative to inverting a Ledoit–Wolf shrinkage estimate of the kind Schäfer and Strimmer [2005] recommend for gene networks. And indeed we see this. The sparsity prior wins clearly where precisions are genuinely sparse or banded (`sparse_precision` 0.63, `hierarchical` 0.68, `ar1` 0.70), ties across the dense elliptope middle. But it loses under `factor` (1.09), and under `residuals` it ties at 1.00 while winning on none of the twenty draws. No mystery here. Dense low-rank structure is the worst case for sparsity.

We also performed a smaller companion study. With oracle shrinkage intensities, the constant-correlation target of Ledoit and Wolf [2004b] beats the scaled-identity target of Ledoit and Wolf [2004a] on every draw under **sparse_precision** (median loss ratio 0.62) and loses on nineteen of twenty draws under **residuals**. Even the choice between two targets from the same authors is ensemble-relative!

Turning to other types of tests next, we have to switch to measuring hypothesis tests, rates and counts. See (Table 2) which we now discuss item by item.

Evolutionary biology. Cheverud [1996] declares two genetic covariance matrices similar when their mean response-vector correlation under common random selection gradients exceeds the null for independent random vectors. We apply the test to independent pairs drawn from the same ensemble. This is a population property with no sampling noise, so a sound test should declare similarity about 5% of the time. It is at or below nominal under **factor** and **archakov_hansen** (0%) and **animals** (10%), and declares independent matrices similar 100% of the time under eleven of the fifteen ensembles. Cheverud and Marroig [2007] concede as much in passing: in practice the null of no shared structure “is nearly always rejected.” Note however that under **sparse_precision** the mean skews correlation between independent matrices is 0.97, because near-diagonal matrices respond to any skewer with the skewer itself. This observation is not new. Indeed Rohlf [2017] raises this objection in general terms, anticipating the measurement here.

Climatology. North et al. [1982] give a rule of thumb: when an eigenvalue’s sampling error $\lambda\sqrt{2/T}$ is comparable to the spacing to a neighbor, the sampling error in the associated EOF is comparable to the neighboring EOF. They draw the converse as a first-order corollary — a stand-out eigenvalue has small EOF sampling error — while saying plainly that the rule offers no guidance on where to truncate. It is the corollary we score. Where does this work? We score the false-reassurance rate, which is to say that among draws the rule declares separated, we look at how often the leading sample eigenvector is mostly wrong (squared alignment with the truth below one half). The rate is 0% under six ensembles, 57% under **wishart**, and 100% under **sparse_precision**, which declares separation on only 12% of draws but is always wrong when it does. We take from this that the rule is a first-order perturbation statement and excellent for spiked spectra. However one should also expect it to be unreliable exactly where spectra are quasi-degenerate.

Ecology. Guillot and Rousset [2013] report that the Mantel test [Mantel, 1967] rejects independence between distance matrices at 25–55% under strong spatial autocorrelation, at nominal 5%. Our audit puts the original claim and the critique on the same footing (two independent site variables drawn with covariance C with Euclidean distance matrices and 499 label permutations). The realized type-I error is the nominal 5% under **onion** and **vine** (and 6% under **lkj**), and 38–43% under **factor**, **kernel**, **animals**, and **residuals**. This reproduces the published inflation under locality-and-factor structure and shows calibration under dense ellipsope measures. So in keeping with our theme, one could say “the Mantel test is broken” or “the Mantel test is fine” depending on which ensemble one chooses.

Quantitative genetics. Kirkpatrick [2009] reports effective dimension counts $n_D = \sum_i \lambda_i / \lambda_{\max}$ of roughly one to two for published genetic covariance matrices, read as evolutionary constraint.

ensemble	skewers false sim.	North rule false reass.	Mantel type-I	n_D ratio est/true
animals	10%	0%	40%	0.89
ar1	100%	20%	13%	0.54
archakov_hansen	0%	9%	28%	0.78
block_equicorr	100%	0%	15%	0.77
factor	0%	0%	38%	0.90
hierarchical	100%	0%	6%	0.78
kernel	100%	3%	39%	0.88
lkj	100%	26%	6%	0.53
onion	100%	33%	5%	0.57
residuals	95%	0%	43%	0.92
sparse_precision	100%	100%	7%	0.42
spectrum	100%	10%	9%	0.52
vine	100%	33%	5%	0.55
walk	100%	0%	8%	0.83
wishart	100%	57%	7%	0.47

Table 2: Four audited claims about inference. Column 1: share of independent matrix pairs that random skewers declares similar (sound: $\approx 5\%$). Column 2: among draws North’s rule declares separated, share whose leading EOF is mostly wrong (squared alignment below $\frac{1}{2}$). Column 3: realized type-I error of the Mantel test at nominal 5%. Column 4: estimated over true effective dimensions of a covariance matrix at $T = 15$ (honest: near one).

At the low precision typical of breeding designs, $T = 15$ here, the estimated n_D deflates the truth by an ensemble-dependent factor. A true n_D of 12.3 under **sparse_precision** is reported as 5.2 (ratio 0.42), while **factor** and **animals** are nearly honest (0.90, 0.89) simply because their truth is already low-dimensional. Across all fifteen ensembles the estimate never exceeds 5.2, whatever the truth – which should give anyone pause. A low reported n_D cannot by itself distinguish genuinely low-dimensional biology from a noisy estimate of diffuse structure.

7 Do claims scale?

Every study above fixes $n = 30$. A reasonable objection is that ensemble-relativity is a small-panel artifact, that estimation noise dominates at $n = 30$ and would wash out with more assets. So we checked. We re-run the Ledoit–Wolf intensity demonstration of Section 4 and five of the seven Table 1 claims (finance, econophysics, data assimilation, signal processing, and the graphical lasso, both the cross-validated penalty and a fixed one) across a 5×5 grid, $n \in \{30, 100, 300, 1000, 3000\}$ crossed with $T/n \in \{0.5, 1, 2, 4, 8\}$, medians over twenty-four seeded draws per cell (twelve at $n = 1000$ and $n = 3000$, where compute starts to bite). Figure 2 plots all thirty-five cells of every audit, one line per ensemble; the two largest passes alone cost on the order of a hundred worker-hours apiece, so this was not a check we could afford to run casually.

The short answer is that the spread does not close. At $n = 3000$, $T/n = 1$, the Ledoit–Wolf intensity still ranges from 0.002 (**factor**) to 0.780 (**sparse_precision**), if anything a wider ratio than the 0.157–0.720 spread we saw at $n = 30$. Marchenko–Pastur clipping helps **sparse_precision** more as n grows (0.293 $\rightarrow$ 0.148 at $T/n = 2$) and hurts **factor** far more

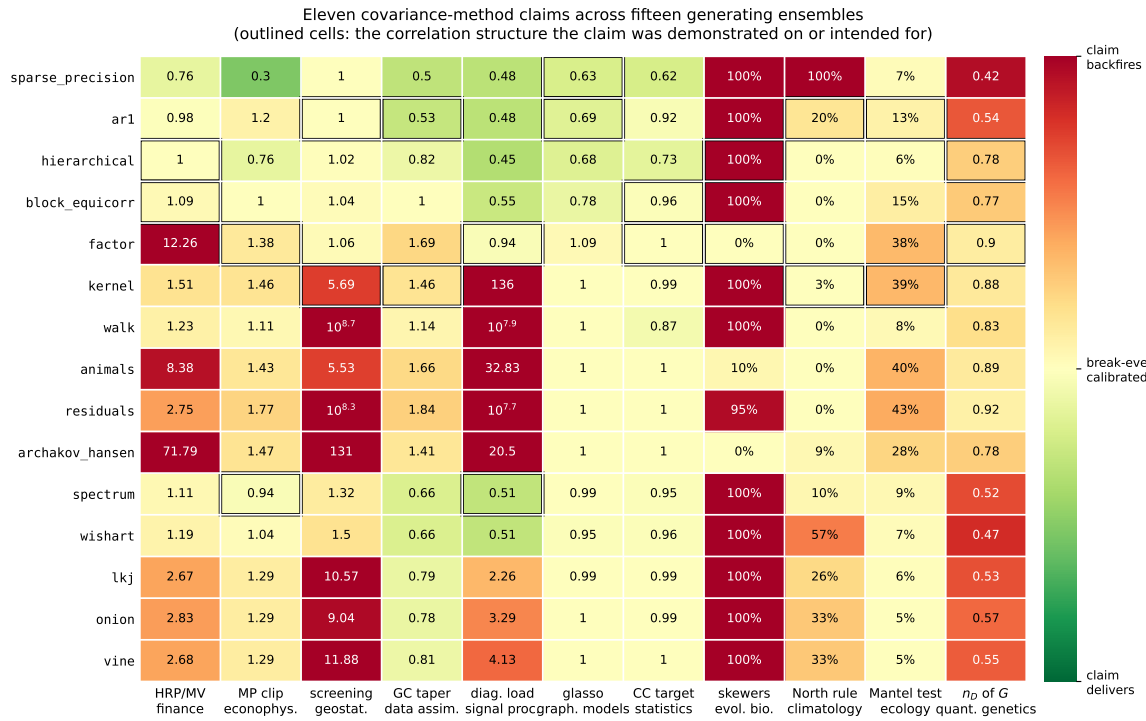


Figure 1: The audit matrix: eleven claims (columns, with their fields) against fifteen generating ensembles (rows, ordered roughly from structured to dense). Green: the claim delivers under that measure. Yellow: break-even or calibrated. Red: the claim backfires or the test is broken. Cell entries are the medians and rates of Tables 1 and 2. Outlined cells mark each claim’s home turf, the correlation structure it was demonstrated on or intended for.

(1.366 $\rightarrow$ 9.432); its verdict does not flip for a single literature ensemble between $n = 30$ and $n = 3000$ at fixed $T/n = 2$, and neither does diagonal loading’s, the graphical lasso’s, or Gaspari–Cohn tapering’s.¹ So for six of the seven claims re-run, a claim’s ensemble-verdict at $n = 30$ turns out to be a reliable predictor of its verdict at a hundred times the dimension.

The exception is hierarchical risk parity, but only under one of the two ensembles that looked exceptional at $n = 30$. At $n = 30$, $T/n = 2$, HRP roughly breaks even against long-only minimum variance under **ar1** and **hierarchical**, the two literature ensembles whose structure the bisection tree is built to match (ratios 0.890, 0.962, twenty-four draws). At $n = 3000$, the twelve-draw grid median moves to 1.259 and 2.580, both apparent losses. That surprised us enough that we ran it back. A targeted replication at forty-eight draws confirms the **hierarchical** loss (median 2.557; every single draw favors minimum variance) and retracts the **ar1** one: the forty-eight-draw median is 0.914, 56% of draws still favor HRP, and the ratio ranges from 0.69 to 1.89.² The twelve-draw estimate had simply landed on the high side of a noisy statistic – a good reminder that a rep count that is fine for most cells is not automatically fine for all of them.

sparse_precision remains an outright HRP win at every scale we tested, and **ar1** remains close to a tie at every scale we tested. Only **hierarchical** genuinely flips, and the cause is the ensemble rather than the algorithm: its cluster correlation is a function of absolute cluster size, so as n grows the matrix thins toward plain equicorrelation — its non-equicorrelation content falls fourfold from $n = 30$ to $n = 3000$, and the share of pairs correlated above 0.2 falls a hundredfold. HRP’s home turf evaporates under it, while the long-only quadratic program keeps exact use of the full covariance regardless of n . So a claim’s home-turf verdict is not guaranteed to survive scale. And, as we just saw, a low rep count is not guaranteed to survive a check either.

So we went looking for more of the same. At $n = 3000$, $T/n = 2$, three literature ensembles sit within 2% of the clip/raw = 1 threshold on the twelve-draw grid: **spectrum** (0.996), **kernel** (1.017), **archakov_hansen** (1.020) – closer calls than HRP’s **ar1** ever was. A forty-eight-draw check confirms all three (medians 0.996, 1.018, 1.021; no range wider than [0.98, 1.06]).³ Nothing moves. The difference from HRP is the statistic itself: a Frobenius-norm ratio has almost no per-draw variance, while HRP’s long-only optimizer clearly does, so twelve draws are trustworthy for one and not for the other. The fragility we found is a property of the audit, not of small rep counts in general.

Five of the fifteen ensembles turn numerically rank-deficient at the largest dimensions tested and need a positive-semidefinite floor before any audit runs: three package-native curiosities (**walk** on 58% of $n = 3000$ panels, **residuals** on 33%, **animals** on 17%) and two literature ensembles (**archakov_hansen** on 14%, **kernel** on 3%). The floor is applied before scoring, so the numbers above already reflect it; a reader citing an **archakov_hansen** or **kernel** verdict at $n = 3000$ should know that a seventh to a thirtieth of its draws needed rescue just to stay a valid correlation matrix.

¹The fixed-penalty graphical lasso records one nominal flip, **1kj** from 1.004 to 1.000, indistinguishable from a tie. The cross-validated graphical lasso is incomplete at $n = 1000$, roughly half its cells missing to per-cell timeouts, and dropped entirely at $n = 3000$, where a single fit runs into the hour.

²**hrp_n3000_replication.py**, beside this paper. Reps 0–11 reuse the grid’s own deterministic seeding; reps 12–47 extend it under the identical scheme.

³**clip_n3000_replication.py**, beside this paper, same seeding scheme as the HRP check.

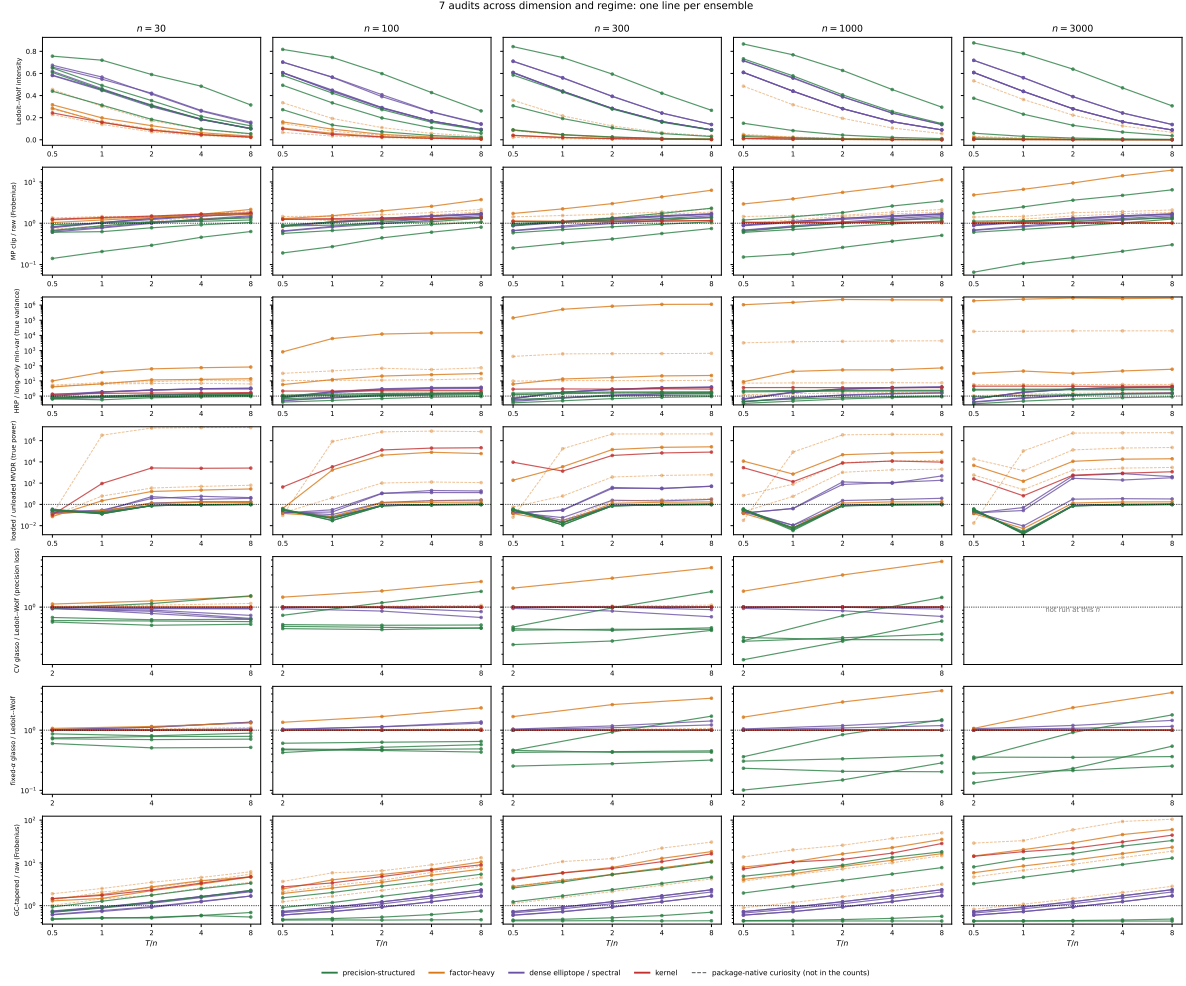


Figure 2: Seven audits (rows: Ledoit–Wolf intensity, MP clipping, HRP, MVDR loading, cross-validated and fixed-penalty graphical lasso, Gaspari–Cohn tapering) across five dimensions (columns, $n = 30$ to 3000) and five regimes within each panel ($T/n = 0.5$ to 8), one line per ensemble, colored by structural family; dashed lines are package-native curiosities. Rows share a y -axis scale within each audit so drift with n is visible. Cross-validated glasso is not run at $n = 3000$.

8 One leaderboard per measure

The audits test one claim at a time; the same harness can just as easily rank whole libraries. We fit the six covariance estimators of scikit-learn and the eleven batch-applicable estimators of **precise** to the same twenty seeded panels per ensemble and rank them by median relative Frobenius loss against the generating covariance (Figure 3). Five estimators take first place somewhere: Ledoit–Wolf on four ensembles, OAS and fixed shrinkage and the graphical lasso on three each, and, perhaps surprisingly, the raw sample covariance on **animals** and **kernel**, the two ensembles the intensity table already marked as wanting almost no shrinkage. The graphical lasso is first on the three precision-structured rows and tenth or worse under **animals**, **residuals**, and **factor** – a big fall. The **precise** estimators are built for time-varying covariance and pay for that design on static panels, filling the lower ranks, which seems fair enough. The upshot is that a library default is really a bet on a measure, whether or not the author meant it that way.

9 Scope

The audits measure the ensemble-sensitivity of a claim, often outside the careful scope of the authors’ original intent. Stein proved that screening depends on smoothness; Bickel and Levina [2008] state the bandable class their banding rates require; López de Prado demonstrated HRP on equity covariances and said so; the data-assimilation literature has always tied localization to physical distance.

The interpretation risk if any usually arises downstream with less fastidious renderings of the original verdict. Those certified under one measure are quoted, cited, and defaulted into software as if they were measure-free, and the second-hand version of a scoped claim is usually unscoped. There is an obvious solution. Naming the ensembles and implementing tests can ameliorate the danger and improve understanding.

10 Remarks

The pattern across Figure 1 is not random: each field’s signature claim performs best on the ensembles that resemble the data the field grew up on, with instructive exceptions where a home-turf cell fails, as when screening fails inside the kernel ensemble. **sparse_precision** is kind to hierarchical bisection, eigenvalue cleaning, localization, and the graphical lasso at once, yet breaks random skewers and North’s rule. The dense elliptope samplers (**lkj**, **onion**, **vine**) are kind to almost nothing except Gaspari–Cohn localization, which delivers on all three (0.78–0.81), and the Mantel test, which they hold at or near its nominal level; they are also the most common default test bed. The same mechanism recurs under different names: the hedge that unconstrained minimum variance exploits is the null-steering that diagonal loading erases.

A survey of prominent covariance-method claims across further fields yields a docket for the next round, kept beside this paper. Sorting it actually teaches a taxonomy of its own. Claims that impose structure on the truth are ensemble-relative; theorems about Wishart sampling noise, on the other hand, hold for every true covariance and simply cannot flip.

The structural claims are naturally the audit targets. Paz and Sánchez [2015] taper clustering-statistic covariances — explicitly declining to assume the far off-diagonal entries

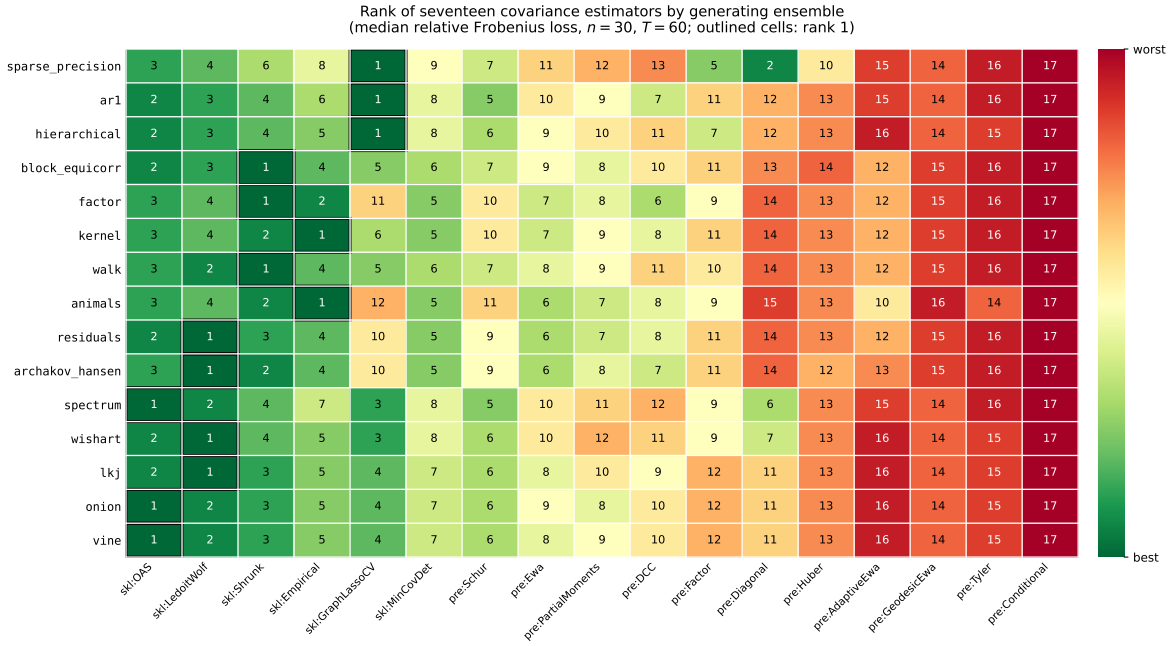


Figure 3: The leaderboard matrix: seventeen covariance estimators, six from scikit-learn and eleven from **precise**, ranked within each generating ensemble by median relative Frobenius loss at $n = 30$, $T = 60$, twenty seeded draws per ensemble. Rank-one cells are outlined; columns are ordered by mean rank.

are negligible, though the taper still presumes the bin coordinate carries the locality; Padmanabhan et al. [2016] estimate sparse cosmological precisions and presume a sparse graph. Zhang and Horvath [2005] select gene-network powers by a scale-free-topology criterion that is itself, whether or not the authors framed it this way, a statement about the generating measure of gene correlation matrices. Smith et al. [2011] demonstrated the win of partial over full correlation for network detection on sparse simulated networks, and report the ordering reversing at fifty nodes. Norberg et al. [2009] judged jackknife covariances for galaxy surveys, and the verdict there hinges on cross-region correlation structure.

A second strand turns on spectral shape instead. Rule N [Preisendorfer, 1988] truncates EOFs by comparing sample eigenvalues against the null spectrum of independent data of the same shape, which is to say against an ensemble. Ledoit and Wolf [2012] show the edge of nonlinear over linear shrinkage growing with spectral dispersion; Fan et al. [2013] build POET on pervasive factors plus sparse residuals; Frank and Friedman [1993] show the PCR-and-PLS-versus-ridge contest turning on whether the true coefficient vector aligns with the high-spread directions of the predictor eigenstructure. Each of these, in our view, carries a predicted flip waiting to be checked.

The invariant theorems serve as null controls, and we lean on them deliberately. Hartlap et al. [2007] debias the inverse sample covariance exactly, for every true covariance; Dodelson and Schneider [2013] derive a parameter-variance inflation that, in the Gaussian limit, depends only on counts; Sellentin and Heavens [2016] marginalize the estimated covariance exactly under Wishart sampling; Lowry et al. [1992] prove the MEWMA chart depends on a known covariance only through a noncentrality parameter. Put simply, an audit pipeline that finds ensemble-dependence in the Hartlap factor has a bug, so we ran that control: with one fixed covariance per ensemble and four hundred panels at $(n, T) = (30, 60)$, the trace of the raw average inverse sample covariance overshoots the true inverse trace by a factor of 2.07 to 2.13 under every ensemble, against the Wishart factor $(T - 1)/(T - n - 2) = 2.11$, and the corrected ratio is 0.98 to 1.01 everywhere. Good – the pipeline reports no ensemble-dependence where none can exist. If anything, the most instructive failures are invariant theorems wrapped in ensemble-relative plug-in practice, as when those corrections are applied to shrunk or tapered estimates whose distribution is no longer Wishart.

There is no free lunch. No method in the tables dominates across the rows. Each owns the ensembles whose structure it assumes, and the matrix draws the borders. We leave to future work the remaining docket entries, beginning with the nonlinear-against-linear shrinkage comparison and the WGCNA scale-free criterion.

References

- Ignacio Alvarez, Jarad Niemi, and Matt Simpson. Bayesian inference for a covariance matrix. In *Proceedings of the 26th Annual Conference on Applied Statistics in Agriculture*, pages 71–82, 2014.
- Ilya Archakov and Peter Reinhard Hansen. A new parametrization of correlation matrices. *Econometrica*, 89(4):1699–1715, 2021.
- Peter J. Bickel and Elizaveta Levina. Regularized estimation of large covariance matrices. *Annals of Statistics*, 36(1):199–227, 2008.

- Anne-Laure Boulesteix, Sabine Lauer, and Manuel J. A. Eugster. A plea for neutral comparison studies in computational sciences. *PLoS ONE*, 8(4):e61562, 2013.
- Blair D. Carlson. Covariance matrix estimation errors and diagonal loading in adaptive arrays. *IEEE Transactions on Aerospace and Electronic Systems*, 24(4):397–401, 1988.
- James M. Cheverud. Quantitative genetic analysis of cranial morphology in the cotton-top (*Saguinus oedipus*) and saddle-back (*S. fuscicollis*) tamarins. *Journal of Evolutionary Biology*, 9(1):5–42, 1996.
- James M. Cheverud and Gabriel Marroig. Comparing covariance matrices: Random skewers method compared to the common principal components model. *Genetics and Molecular Biology*, 30(2):461–469, 2007.
- Henry Cox, Robert M. Zeskind, and Mark M. Owen. Robust adaptive beamforming. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 35(10):1365–1376, 1987.
- Scott Dodelson and Michael D. Schneider. The effect of covariance estimator error on cosmological parameter constraints. *Physical Review D*, 88(6):063537, 2013.
- Jianqing Fan, Yuan Liao, and Martina Mincheva. Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B*, 75(4):603–680, 2013.
- Ildiko E. Frank and Jerome H. Friedman. A statistical view of some chemometrics regression tools. *Technometrics*, 35(2):109–135, 1993.
- Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441, 2008.
- Gregory Gaspari and Stephen E. Cohn. Construction of correlation functions in two and three dimensions. *Quarterly Journal of the Royal Meteorological Society*, 125(554):723–757, 1999.
- Louis W. Glorfeld. An improvement on Horn’s parallel analysis methodology for selecting the correct number of factors to retain. *Educational and Psychological Measurement*, 55(3):377–393, 1995.
- Gilles Guillot and François Rousset. Dismantling the Mantel tests. *Methods in Ecology and Evolution*, 4(4):336–344, 2013.
- Thomas M. Hamill, Jeffrey S. Whitaker, and Chris Snyder. Distance-dependent filtering of background error covariance estimates in an ensemble Kalman filter. *Monthly Weather Review*, 129(11):2776–2790, 2001.
- Johanna Hardin, Stephan Ramon Garcia, and David Golan. A method for generating realistic correlation matrices. *Annals of Applied Statistics*, 7(3):1733–1762, 2013.
- Jan Hartlap, Patrick Simon, and Peter Schneider. Why your model parameter confidences might be too optimistic: Unbiased estimation of the inverse covariance matrix. *Astronomy & Astrophysics*, 464(1):399–404, 2007.
- J. N. Hooker. Testing heuristics: We have it all wrong. *Journal of Heuristics*, 1(1):33–42, 1995.

- John L. Horn. A rationale and test for the number of factors in factor analysis. *Psychometrika*, 30(2):179–185, 1965.
- P. L. Houtekamer and Herschel L. Mitchell. A sequential ensemble Kalman filter for atmospheric data assimilation. *Monthly Weather Review*, 129(1):123–137, 2001.
- Harry Joe. Generating random correlation matrices based on partial correlations. *Journal of Multivariate Analysis*, 97(10):2177–2189, 2006.
- Henry F. Kaiser. The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20(1):141–151, 1960.
- Cari G. Kaufman, Mark J. Schervish, and Douglas W. Nychka. Covariance tapering for likelihood-based estimation in large spatial data sets. *Journal of the American Statistical Association*, 103(484):1545–1555, 2008.
- Mark Kirkpatrick. Patterns of quantitative genetic variation in multiple dimensions. *Genetica*, 136(2):271–284, 2009.
- Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999.
- Laurent Laloux, Pierre Cizeau, Marc Potters, and Jean-Philippe Bouchaud. Random matrix theory and financial correlations. *International Journal of Theoretical and Applied Finance*, 3(3):391–397, 2000.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004a.
- Olivier Ledoit and Michael Wolf. Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4):110–119, 2004b.
- Olivier Ledoit and Michael Wolf. Nonlinear shrinkage estimation of large-dimensional covariance matrices. *Annals of Statistics*, 40(2):1024–1060, 2012.
- Daniel Lewandowski, Dorota Kurowicka, and Harry Joe. Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis*, 100(9):1989–2001, 2009.
- Marcos López de Prado. Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4):59–69, 2016.
- Cynthia A. Lowry, William H. Woodall, Charles W. Champ, and Steven E. Rigdon. A multivariate exponentially weighted moving average control chart. *Technometrics*, 34(1):46–53, 1992.
- Nathan Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2):209–220, 1967.
- Gautier Marti. CorrGAN: Sampling realistic financial correlation matrices using generative adversarial networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8459–8463, 2020.

- Gautier Marti, Victor Goubet, and Frank Nielsen. cCorrGAN: Conditional correlation GAN for learning empirical conditional distributions in the elliptope. In *Geometric Science of Information*, Lecture Notes in Computer Science, pages 613–620, 2021.
- Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019.
- Christina Nießl, Moritz Herrmann, Chiara Wiedemann, Giuseppe Casalicchio, and Anne-Laure Boulesteix. Over-optimism in benchmark studies and the multiplicity of design and analysis options when interpreting their results. *WIREs Data Mining and Knowledge Discovery*, 12(2):e1441, 2022.
- Peder Norberg, Carlton M. Baugh, Enrique Gaztañaga, and Darren J. Croton. Statistical analysis of galaxy surveys – I. robust error estimation for two-point clustering statistics. *Monthly Notices of the Royal Astronomical Society*, 396(1):19–38, 2009.
- Gerald R. North, Thomas L. Bell, Robert F. Cahalan, and Fanthune J. Moeng. Sampling errors in the estimation of empirical orthogonal functions. *Monthly Weather Review*, 110(7):699–706, 1982.
- Nikhil Padmanabhan, Martin White, Harrison H. Zhou, and Ross O’Connell. Estimating sparse precision matrices. *Monthly Notices of the Royal Astronomical Society*, 460(2):1567–1576, 2016.
- Jochen Papenbrock, Peter Schwendner, Markus Jaeger, and Stephan Krügel. Matrix evolutions: Synthetic correlations and explainable machine learning for constructing robust investment portfolios. *Journal of Financial Data Science*, 3(2):51–69, 2021.
- Dante J. Paz and Ariel G. Sánchez. Improving the precision matrix for precision cosmology. *Monthly Notices of the Royal Astronomical Society*, 454(4):4326–4334, 2015.
- Rudolph W. Preisendorfer. *Principal Component Analysis in Meteorology and Oceanography*, volume 17 of *Developments in Atmospheric Science*. Elsevier, Amsterdam, 1988.
- F. James Rohlf. The method of random skewers. *Evolutionary Biology*, 44(4):542–550, 2017.
- Juliane Schäfer and Korbinian Strimmer. A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology*, 4(1):32, 2005.
- Elena Sellentin and Alan F. Heavens. Parameter inference with estimated covariance matrices. *Monthly Notices of the Royal Astronomical Society: Letters*, 456(1):L132–L136, 2016.
- Stephen M. Smith, Karla L. Miller, Gholamreza Salimi-Khorshidi, Matthew Webster, Christian F. Beckmann, Thomas E. Nichols, Joseph D. Ramsey, and Mark W. Woolrich. Network modelling methods for FMRI. *NeuroImage*, 54(2):875–891, 2011.
- Michael L. Stein. The screening effect in kriging. *Annals of Statistics*, 30(1):298–323, 2002.
- David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.

Bin Zhang and Steve Horvath. A general framework for weighted gene co-expression network analysis. *Statistical Applications in Genetics and Molecular Biology*, 4(1):17, 2005.